Unit 4 Guided Notes — Geometry: Angles & Triangles

ANSWER KEY

Name the relationship, write the equation, solve for x. Angles always add up to something.

🎯 By the end of this unit, I can…

- Use angle relationships formed by parallel lines and a transversal
- Apply the Triangle Angle-Sum and Exterior-Angle Theorems
- Solve for unknown angles by writing and solving equations

Standard: NC.8.G.5

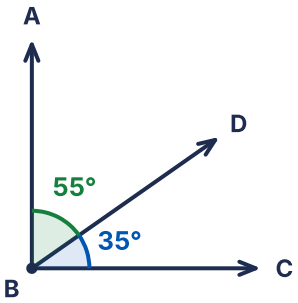
Answer key: every blank and final answer is filled in green.

1 Angle Relationships

Relationship	Definition	Sum / Rule
Complementary	Two angles that add to **90** °	**a + b = 90°**
Supplementary	Two angles that add to **180** °	**a + b = 180°**
Vertical Angles	**Opposite angles when two lines cross**	**Always EQUAL**
Linear Pair	**Adjacent angles on a straight line**	**Add to 180°**

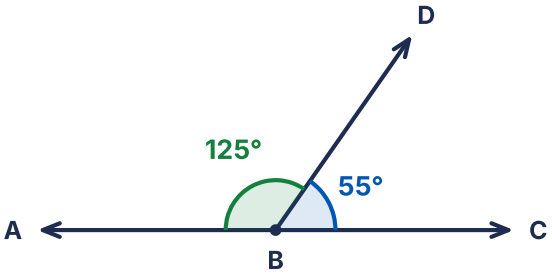
Each angle is named with three letters — the vertex is the MIDDLE letter (e.g. ∠ABD has vertex B). Arcs show which angle each measure belongs to.

COMPLEMENTARY = 90°



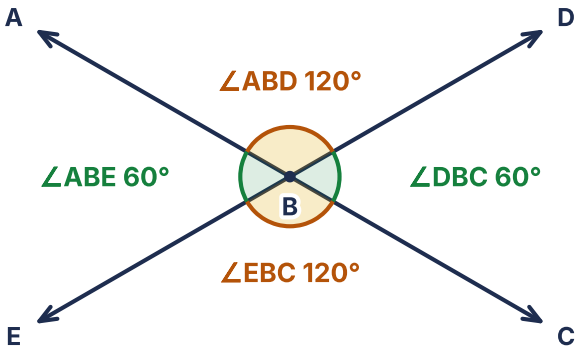
35° + 55° = 90°

SUPPLEMENTARY = 180°



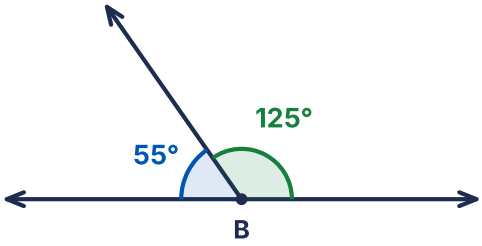
55° + 125° = 180°

VERTICAL ANGLES



Opposite angles are EQUAL: ∠ABD = ∠EBC, ∠DBC = ∠ABE

LINEAR PAIR = 180°



side by side on a line: 125° + 55° = 180°

Quick Check

MC Two angles form a linear pair. One measures 115°. What is the other?

- ● **65°**
- ○ 75°
- ○ 245°

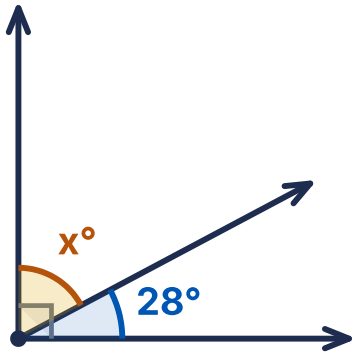
Match the Relationship to its Rule

MATCH **Write the letter of the matching rule on each line.**

B	1. Complementary	**A.**	Always EQUAL
D	2. Supplementary	**B.**	Add to 90°
A	3. Vertical Angles	**C.**	Adjacent angles on a line → 180°
C	4. Linear Pair	**D.**	Add to 180°

✏ Practice: Angle Relationships

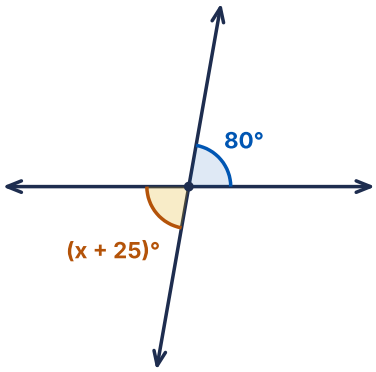
1. ★ Try it The two angles are complementary. Find x.



Answer: 62

How to do it: Complementary angles add to 90°. x + 28 = 90, so x = 90 − 28 = 62.

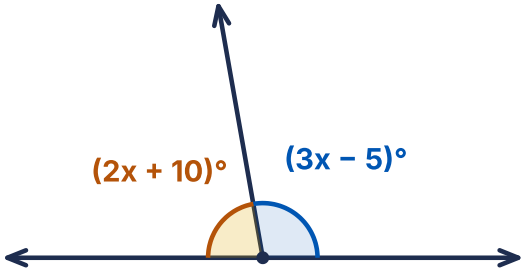
2. ★★ On level Two lines cross. Find x.



Answer: 55

How to do it: The angles are across from each other, so they are vertical angles. Vertical angles are equal: x + 25 = 80. Subtract 25 from both sides: x = 55.

3. ★★★ Challenge The two angles form a linear pair. Find x.

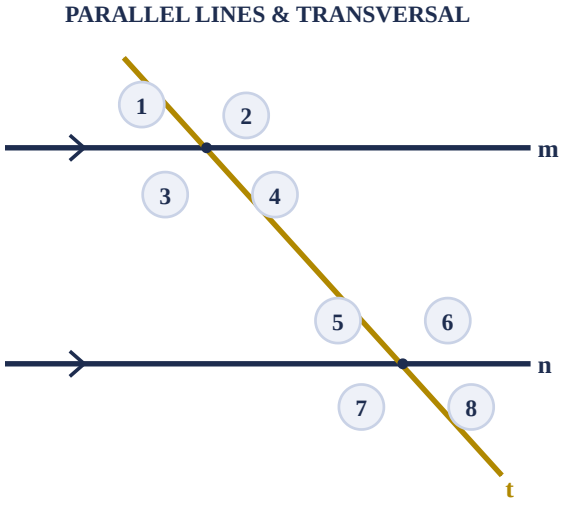


Answer: 35

How to do it: A linear pair adds to 180°. (3x − 5) + (2x + 10) = 180. Combine like terms: 5x + 5 = 180. Subtract 5: 5x = 175. Divide by 5: x = 35. Check: 3(35) − 5 = 100° and 2(35) + 10 = 80°, and 100° + 80° = 180° ✓

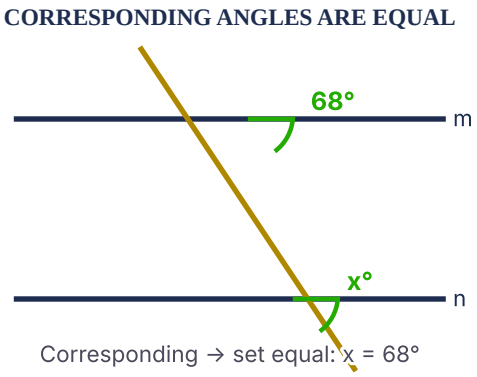
2 Parallel Lines & Transversals

📌 **REMEMBER:** Corresponding, Alternate Interior, Alternate Exterior, and Vertical angles are CONGRUENT (equal). Same-Side Interior and Same-Side Exterior angles are SUPPLEMENTARY (add to 180°).



Two parallel lines (m ∥ n) cut by transversal t form 8 angles. Corresponding: (1,5)(2,6)(3,7)(4,8) · Alt. Interior: (3,6)(4,5) · Alt. Exterior: (1,8)(2,7) · Same-Side Interior: (3,5)(4,6)

Angle Pair Type	Relationship	Rule to Use	Example
Corresponding	Congruent	Set equal	68° → x = 68°
Alternate Interior	Congruent	Set equal	101° → x = 101°
Alternate Exterior	Congruent	Set equal	142° → x = 142°
Vertical	Congruent	Set equal	120° → x = 120°
Same-Side Interior	Supplementary	180 − given	134° → x = 46°
Same-Side Exterior	Supplementary	180 − given	77° → x = 103°



Worked Example — ∠1 = 3x + 15 and ∠2 = 5x − 7 (alternate interior angles)

Alternate interior angles are EQUAL: $3x + 15 = 5x - 7$

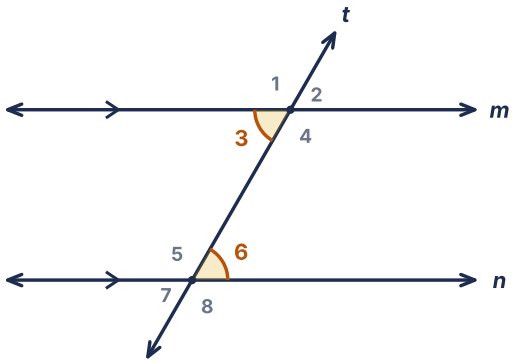
Subtract 3x: $15 = 2x - 7$

Add 7: $22 = 2x$

✓ Divide by 2: $x = 11 \rightarrow$ angle $= 3(11) + 15 = 48°$

✏ Practice: Parallel Lines & Transversals

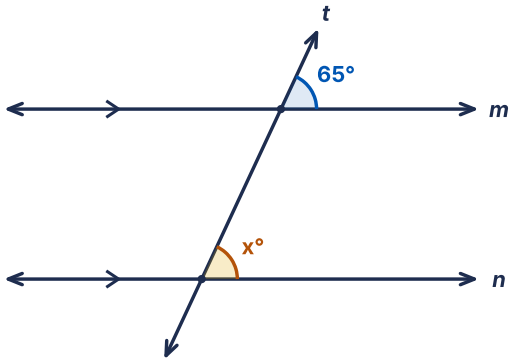
1. ★ Try it Lines m and n are parallel. What kind of angle pair are ∠3 and ∠6?



- ○ Corresponding
- ● **Alternate interior**
- ○ Same-side interior

> **How to do it:** ∠3 and ∠6 are both between the parallel lines (interior). They are on opposite sides of t (alternate). So they are alternate interior angles, and they are equal.

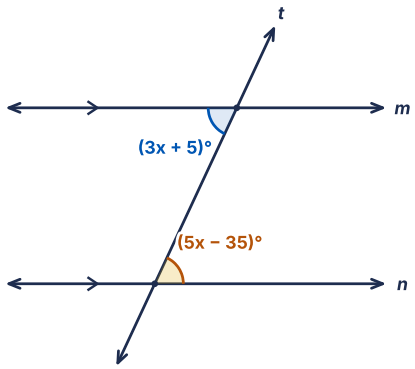
2. ★ Try it Lines m and n are parallel. Find x.



Answer: 65

> **How to do it:** The 65° angle and x are in the same corner at each crossing, so they are corresponding angles. Corresponding angles are equal: x = 65.

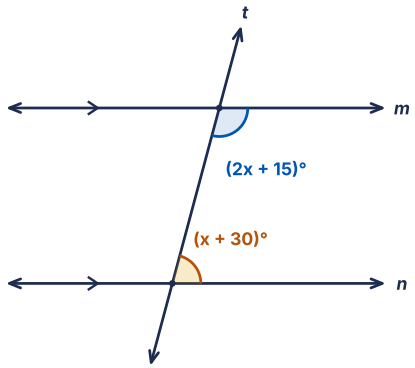
3. ★★ On level Lines m and n are parallel. Find x.



Answer: 20

How to do it: Alternate interior angles are equal: 3x + 5 = 5x − 35. Subtract 3x: 5 = 2x − 35. Add 35: 40 = 2x. Divide by 2: x = 20. Check: 3(20) + 5 = 65° and 5(20) − 35 = 65° ✓

4. ★★★ Challenge Lines m and n are parallel. Find x. Then find the measure of the angle marked (2x + 15)°.



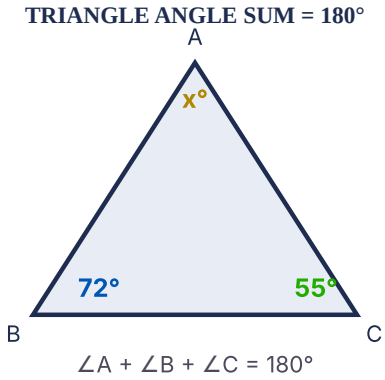
Answer: 105

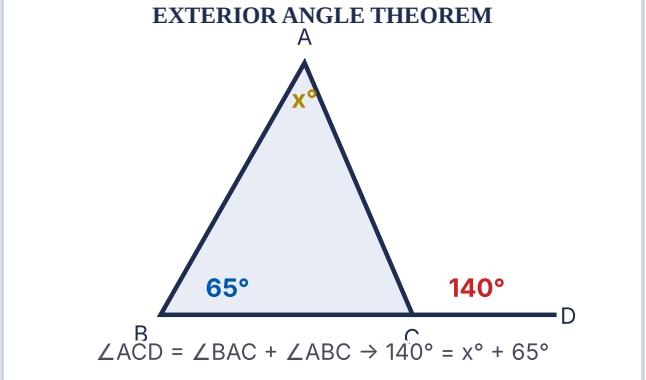
How to do it: Both angles are between the lines on the same side of t, so they are same-side interior angles. They add to 180°: (2x + 15) + (x + 30) = 180 → 3x + 45 = 180 → 3x = 135 → x = 45. The angle is 2(45) + 15 = 105°. Check: the other angle is 45 + 30 = 75°, and 105° + 75° = 180° ✓

3 Triangle Theorems

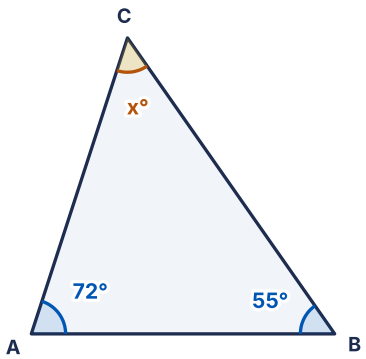
🔑 **TRIANGLE ANGLE SUM:** The three interior angles of a triangle always add up to 180°.

🔑 **EXTERIOR ANGLE THEOREM:** An exterior angle equals the SUM of the two non-adjacent (remote) interior angles.



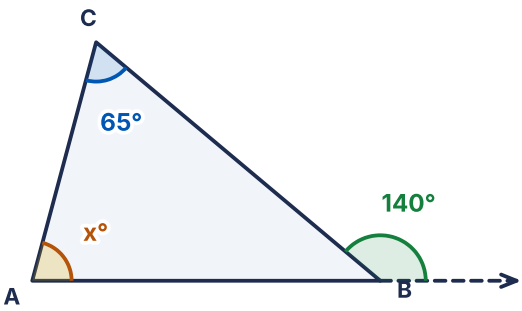


Problem 1 — Triangle Angle Sum: x + 72 + 55 = 180



Solve: x + 127 = 180 → x = 53°

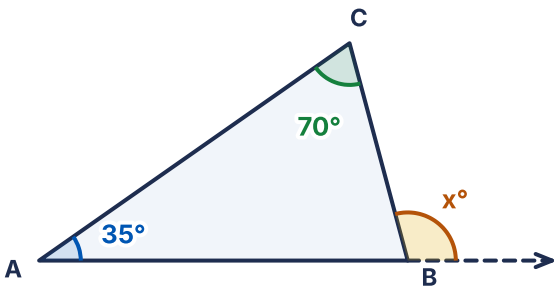
Problem 2 — Exterior Angle Theorem: x + 65 = 140



the outside angle = the two far inside angles

Solve: x + 65 = 140 → x = 75°

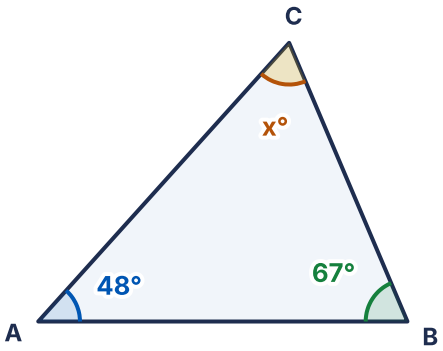
ERROR ANALYSIS Jordan's Exterior Angle Mistake — A triangle has interior angles 35° and 70°. Exterior angle = x. Jordan wrote `x = 70 − 35 = 35°`.



Jordan subtracted instead of adding. The exterior angle = SUM of the remote interior angles: x = 35 + 70 = 105°.

✏ Practice: Triangle Theorems

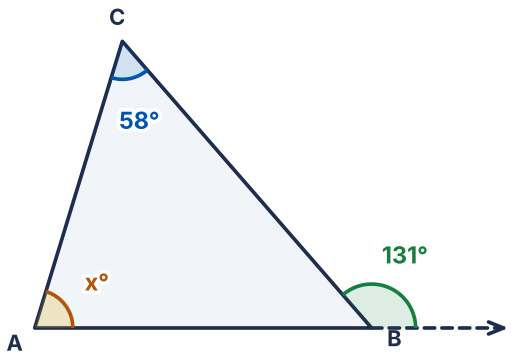
1. ★ Try it Find x.



Answer: 65

How to do it: The angles of a triangle add to 180°. x + 48 + 67 = 180 → x + 115 = 180 → x = 65.

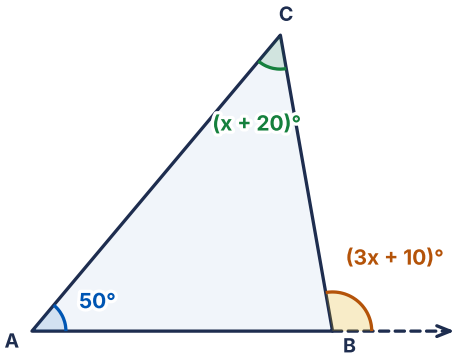
2. ★★ On level The exterior (outside) angle is 131°. Find x.



Answer: 73

How to do it: Exterior Angle Theorem: the outside angle = the sum of the two far (remote) inside angles. x + 58 = 131. Subtract 58: x = 73.

3. ★★★ Challenge Find x. Then find the measure of the exterior angle, (3x + 10)°.



Answer: 100

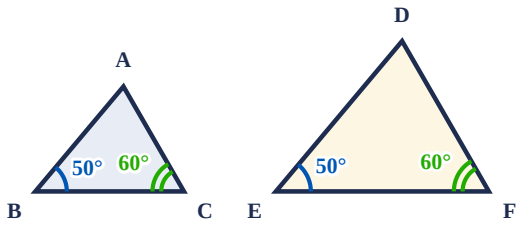
How to do it: Exterior angle = sum of the two far inside angles: 3x + 10 = 50 + (x + 20). Simplify: 3x + 10 = x + 70. Subtract x: 2x + 10 = 70. Subtract 10: 2x = 60, so x = 30. Exterior angle = 3(30) + 10 = 100°. Check: 50 + (30 + 20) = 100 ✓

4 Angle-Angle (AA) Similarity

🔑 **SIMILAR TRIANGLES:** Two triangles are *similar* (written △ABC ~ △DEF) when they have the same shape but not necessarily the same size. Corresponding angles are equal and corresponding sides are in proportion.

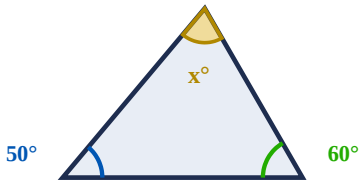
🔑 **ANGLE-ANGLE (AA) CRITERION:** If two angles of one triangle are equal to two angles of another triangle, then the triangles are similar. You only ever need to check TWO pairs of angles.

ANGLE-ANGLE (AA) CRITERION



matching marks = equal angles → △ABC ~ △DEF

WHY TWO ANGLES ARE ENOUGH

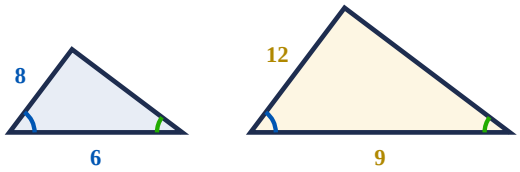


50° + 60° + x° = 180° → x = 70°

the third angle is forced, so all three match

🔑 **WHY TWO ANGLES ARE ENOUGH:** Every triangle's angles add to 180°. Once two pairs of angles match, the third pair has no choice but to match as well — so all three pairs are equal.

SIMILAR → SIDES IN PROPORTION

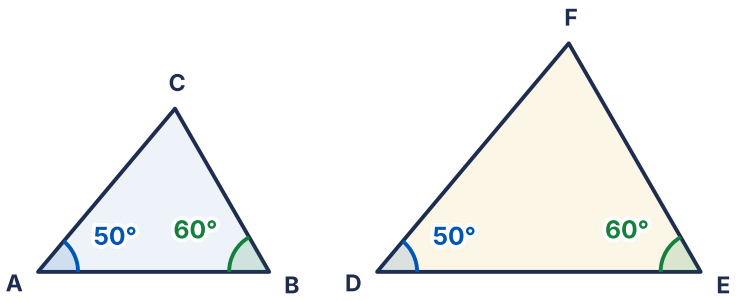


9 ÷ 6 = 1.5 and 12 ÷ 8 = 1.5

scale factor 1.5

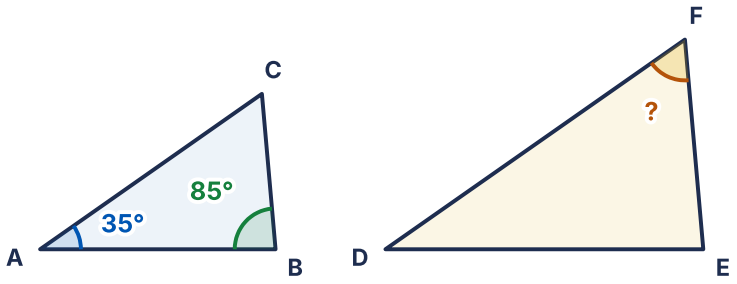
📌 **WATCH OUT:** AA needs two pairs of angles that *match each other* — not simply two angles given in each triangle.

Problem 1 — Are they similar? ∠A = 50°, ∠B = 60° and ∠D = 50°, ∠E = 60°



Decide: Two pairs of angles are equal (50° = 50° and 60° = 60°), so by AA the triangles are similar: △ABC ~ △DEF.

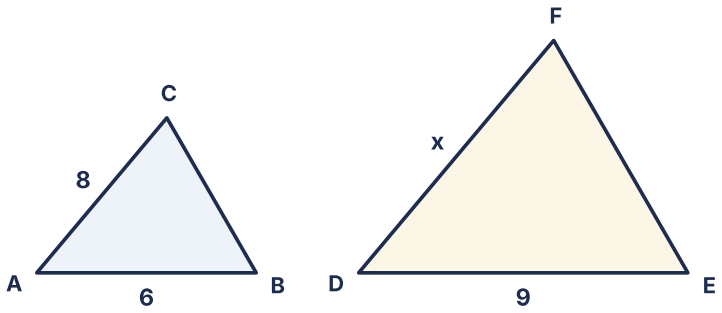
Problem 2 — △ABC ~ △DEF with ∠A = 35° and ∠B = 85°. Find ∠F.



F matches C

Solve: ∠C = 180 − 35 − 85 = 60°. ∠F corresponds to ∠C, so ∠F = 60°.

Problem 3 — Similar triangles: sides 6 and 8 match sides 9 and x. Find x.

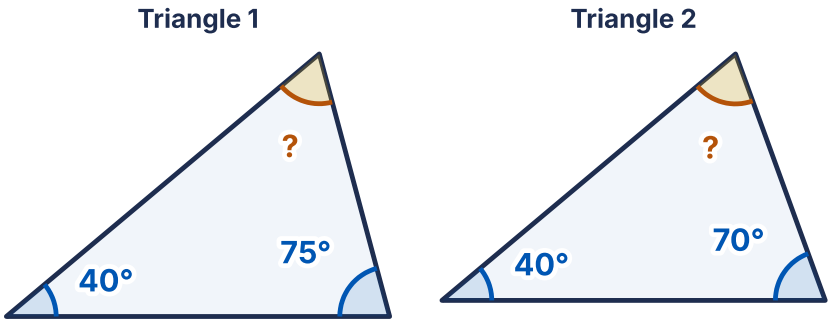


6 → 9 and 8 → x

Solve: Scale factor = 9 ÷ 6 = 1.5, so x = 8 × 1.5 = 12.

ERROR ANALYSIS **Riley's AA Mistake — Triangle 1 has angles 40° and 75°. Triangle 2 has angles 40° and 70°. Riley wrote**

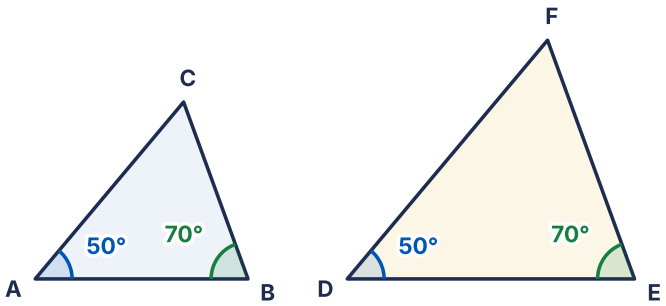
> **"Each triangle shows two angles, so they are similar by AA."**



AA needs two pairs of angles that match *each other*. Only the 40° angles match here. Triangle 1's angles are 40°, 75°, 65°; Triangle 2's are 40°, 70°, 70°. The two sets are different, so the triangles are **not** similar.

✏ Practice: AA Similarity

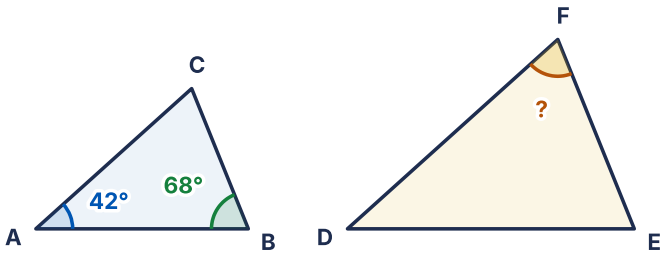
1. ★ Try it Are these two triangles similar?



- ● **Yes — two pairs of angles match**
- ○ No — they are different sizes
- ○ You can't tell

How to do it: ∠A = ∠D = 50° and ∠B = ∠E = 70°. Two pairs of angles match, so the triangles are similar by AA. Size does not matter.

2. ★★ On level △ABC ~ △DEF. Find the measure of ∠F.

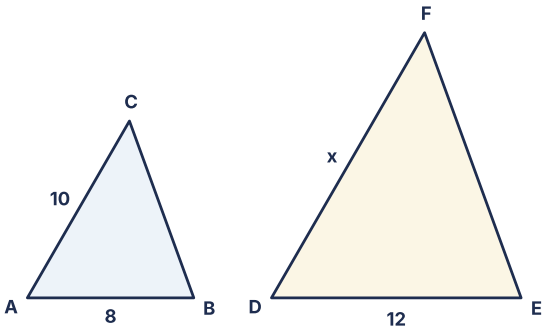


F matches C

Answer: 70

How to do it: ∠C = 180 − 42 − 68 = 70°. ∠F corresponds to ∠C, so ∠F = 70°.

3. ★★ On level The triangles are similar. Side 8 matches side 12, and side 10 matches side x. Find x.



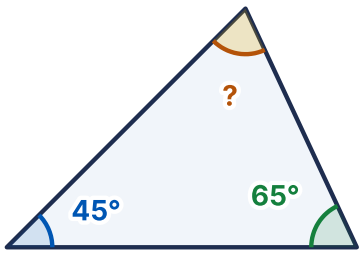
8 → 12 and 10 → x

Answer: 15

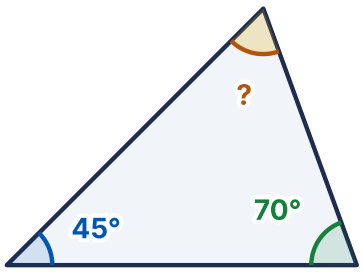
How to do it: Scale factor = 12 ÷ 8 = 1.5. So x = 10 × 1.5 = 15.

4. ★★★ Challenge Triangle 1 shows 45° and 65°. Triangle 2 shows 45° and 70°. Sam says they are NOT similar. Is Sam right?

Triangle 1



Triangle 2



○ Yes — they are not similar

● **No — they are similar**

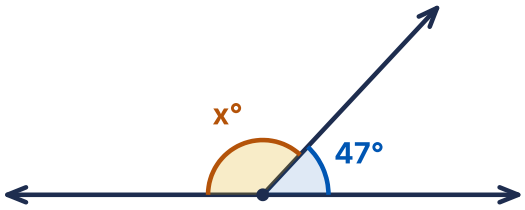
How to do it: Triangle 1: third angle = 180 − 45 − 65 = 70°, so its angles are 45°, 65°, 70°. Triangle 2: third angle = 180 − 45 − 70 = 65°, so its angles are 45°, 70°, 65°. Both have 45°, 65° and 70°. Two pairs match, so they ARE similar by AA. Sam is not right.

✏ Part 2 — Comprehensive Practice

📌 **Show every step on its own line.** Box your final answer.

✏ Comprehensive Practice All sections

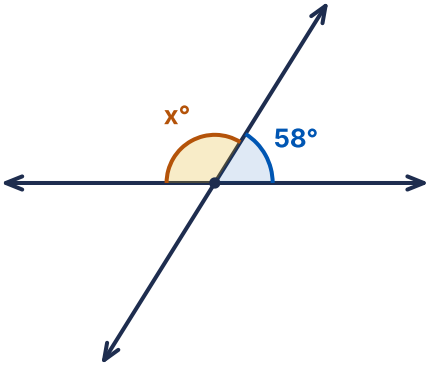
1. ★ Try it The two angles are supplementary. Find x.



Answer: 133

How to do it: Supplementary angles add to 180°. x + 47 = 180, so x = 180 − 47 = 133.

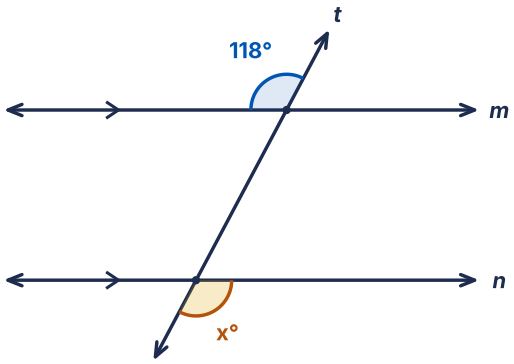
2. ★★ On level Two lines cross. Find x.



Answer: 122

How to do it: x and 58° sit side by side on a straight line, so they are a linear pair. x + 58 = 180, so x = 122. (They are not vertical angles — vertical angles are across from each other.)

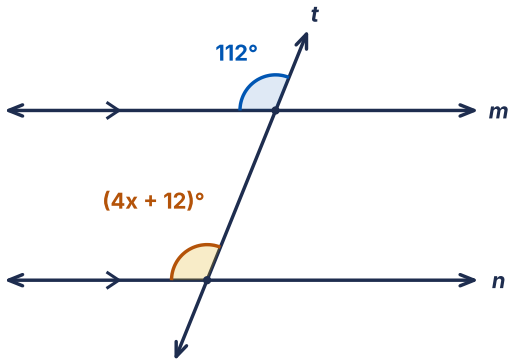
3. ★ Try it Lines m and n are parallel. Find x.



Answer: 118

How to do it: The angles are outside the parallel lines and on opposite sides of t, so they are alternate exterior angles. They are equal: x = 118.

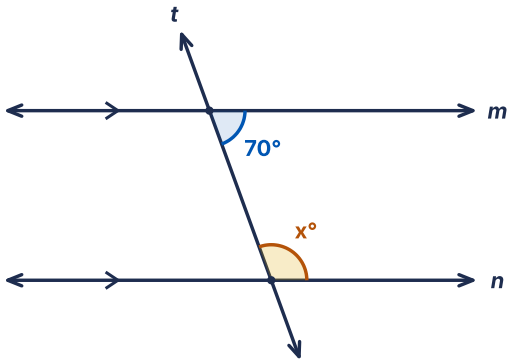
4. ★★ On level Lines m and n are parallel. Find x.



Answer: 25

How to do it: The angles are in the same corner at each crossing, so they are corresponding angles and are equal. 4x + 12 = 112. Subtract 12: 4x = 100. Divide by 4: x = 25.

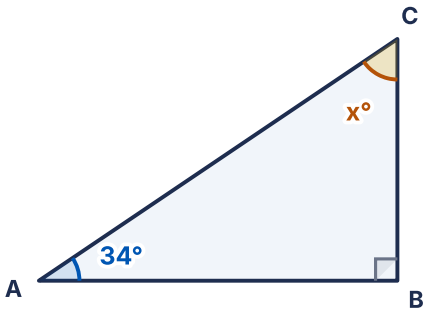
5. ★★★ Challenge Lines m and n are parallel. Maya says x = 70 because the angles "look the same." Find the correct value of x.



Answer: 110

How to do it: The angles are between the lines and on the same side of t, so they are same-side interior angles. Those add to 180°, they are not equal. x + 70 = 180, so x = 110. Maya's mistake: she set them equal.

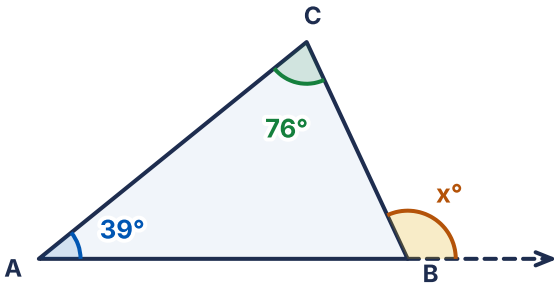
6. ★ Try it Find x. (The square corner is a right angle.)



Answer: 56

How to do it: A right angle is 90°. x + 34 + 90 = 180 → x + 124 = 180 → x = 56.

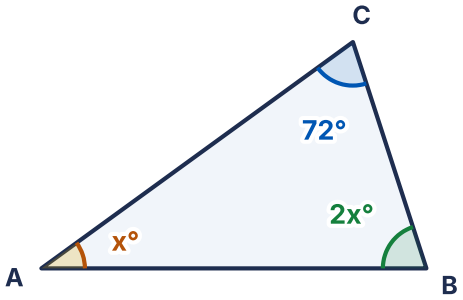
7. ★ Try it Find the exterior angle x.



Answer: 115

How to do it: Exterior Angle Theorem: x = 39 + 76 = 115.

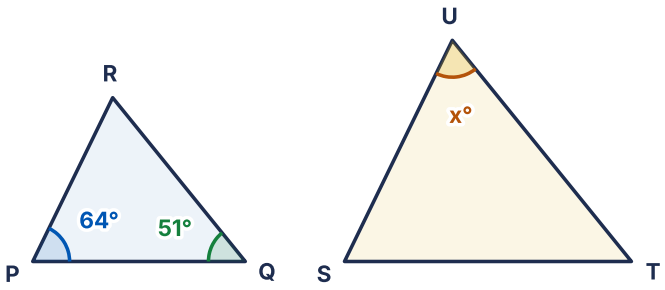
8. ★★★ Challenge Find x.



Answer: 36

How to do it: The angles add to 180°: x + 2x + 72 = 180. Combine like terms: 3x + 72 = 180. Subtract 72: 3x = 108. Divide by 3: x = 36. Check: 36 + 72 + 72 = 180 ✓

9. ★★ On level △PQR ~ △STU. Find x.

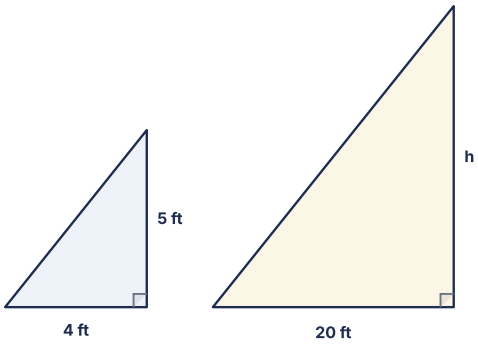


U matches R

Answer: 65

How to do it: ∠R = 180 − 64 − 51 = 65°. ∠U corresponds to ∠R, so x = 65.

10. ★★★ Challenge A 5 ft student casts a 4 ft shadow. At the same time, a tree casts a 20 ft shadow. The two triangles are similar (AA). How tall is the tree?



student and shadow → tree and shadow (not to scale)

Answer: 25

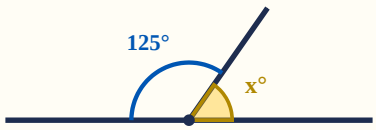
How to do it: The sun hits both at the same angle, and both stand straight up (90°), so the triangles are similar by AA. Scale factor = 20 ÷ 4 = 5. Tree height h = 5 × 5 = 25 ft.

More Practice by Topic

Extra skill practice for each topic. ★ = Try it, ★★ = On level.

A. Angle Relationships — Solve for x

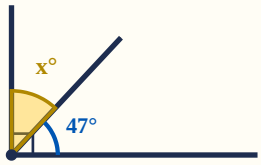
★ Try it **1) Supplementary Angles, given 125°. x = ?**



x = 55°

How to do it: Supplementary angles add to 180°: x = 180 − 125 = 55.

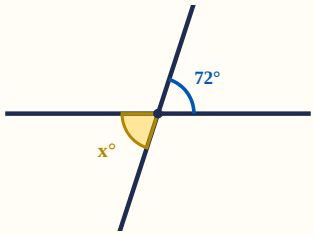
★ Try it **2) Complementary Angles, given 47°. x = ?**



x = 43°

How to do it: Complementary angles add to 90°: x = 90 − 47 = 43.

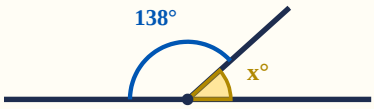
★ Try it **3) Vertical Angles, given 72°. x = ?**



x = 72°

How to do it: Vertical angles are equal: x = 72.

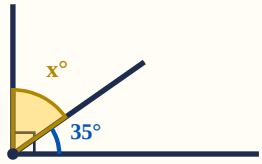
★ Try it **4) Linear Pair, given 138°. x = ?**



x = 42°

How to do it: A linear pair adds to 180°: x = 180 − 138 = 42.

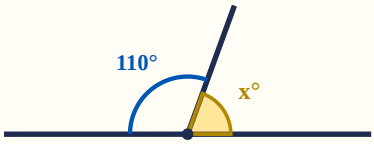
★ Try it **5) Complementary Angles, given 35°. x = ?**



x = **55°**

How to do it: Complementary angles add to 90°: x = 90 − 35 = 55.

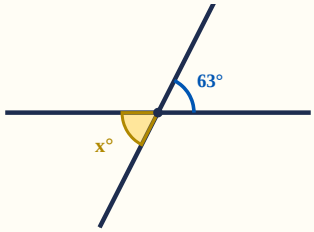
★ Try it **6) Supplementary Angles, given 110°. x = ?**



x = **70°**

How to do it: Supplementary angles add to 180°: x = 180 − 110 = 70.

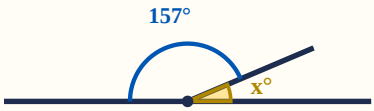
★ Try it **7) Vertical Angles, given 63°. x = ?**



x = **63°**

How to do it: Vertical angles are equal: x = 63.

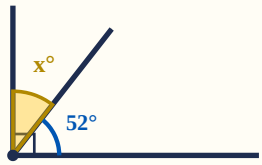
★ Try it **8) Linear Pair, given 157°. x = ?**



x = **23°**

How to do it: A linear pair adds to 180°: x = 180 − 157 = 23.

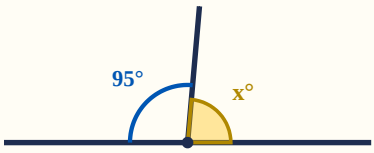
★ Try it **9) Complementary Angles, given 52°. x = ?**



x = **38°**

How to do it: Complementary angles add to 90°: x = 90 − 52 = 38.

★ Try it **10) Supplementary Angles, given 95°. x = ?**

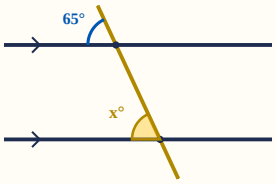


x = **85°**

How to do it: Supplementary angles add to 180°: x = 180 − 95 = 85.

B. Parallel Lines & Transversals

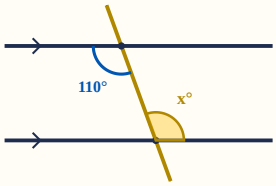
★ Try it **1) Corresponding, given 65°. x = ?**



x = 65°

How to do it: Corresponding angles are equal: x = 65.

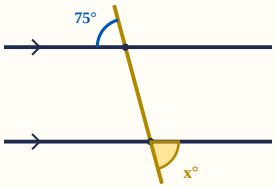
★ Try it **2) Alternate Interior, given 110°. x = ?**



x = 110°

How to do it: Alternate Interior angles are equal: x = 110.

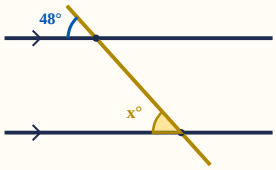
★ Try it **3) Alternate Exterior, given 75°. x = ?**



x = 75°

How to do it: Alternate Exterior angles are equal: x = 75.

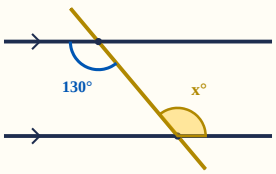
★ Try it **4) Corresponding, given 48°. x = ?**



x = 48°

How to do it: Corresponding angles are equal: x = 48.

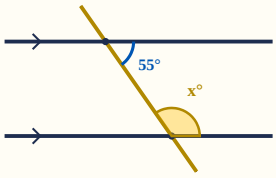
★ Try it **5) Alternate Interior, given 130°. x = ?**



x = 130°

How to do it: Alternate Interior angles are equal: x = 130.

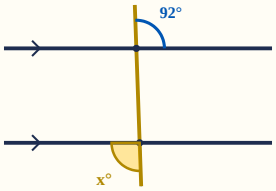
★ Try it **6) Co-Interior, given 55°. x = ?**



x = 125°

How to do it: Co-interior (same-side interior) angles add to 180°: x = 180 − 55 = 125.

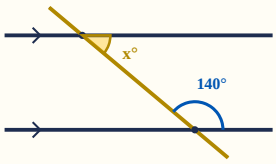
★ Try it **7) Alternate Exterior, given 92°. x = ?**



x = 92°

How to do it: Alternate Exterior angles are equal: x = 92.

★ Try it **8) Co-Interior, given 140°. x = ?**

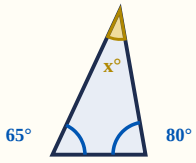


x = 40°

How to do it: Co-interior (same-side interior) angles add to 180°: x = 180 − 140 = 40.

C. Triangle Theorems

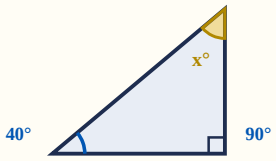
★ Try it **1) Angle Sum, given 65°, 80°. x = ?**



x = **35°**

How to do it: The angles of a triangle add to 180°: x = 180 − 65 − 80 = 35.

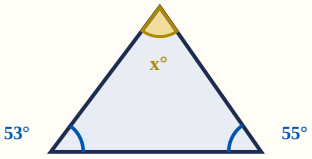
★ Try it **2) Angle Sum, given 40°, 90°. x = ?**



x = **50°**

How to do it: The angles of a triangle add to 180°: x = 180 − 40 − 90 = 50.

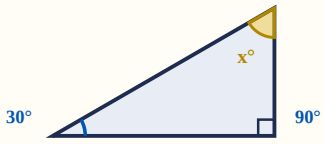
★ Try it **3) Angle Sum, given 53°, 55°. x = ?**



x = **72°**

How to do it: The angles of a triangle add to 180°: x = 180 − 53 − 55 = 72.

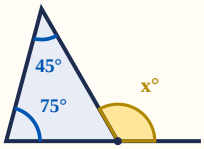
★ Try it **4) Angle Sum, given 30°, 90°. x = ?**



x = **60°**

How to do it: The angles of a triangle add to 180°: x = 180 − 30 − 90 = 60.

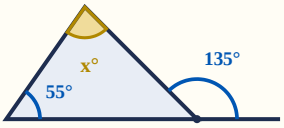
★ Try it **5) Exterior Angle Theorem, remotes 45°, 75°. Exterior x = ?**



x = 120°

How to do it: Exterior angle = sum of the two far inside angles: x = 45 + 75 = 120.

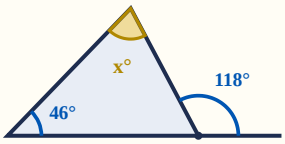
★ Try it **6) Exterior Angle Theorem, remote 55°, exterior = 135°. x = ?**



x = 80°

How to do it: Exterior angle = sum of the two far inside angles: x + 55 = 135, so x = 135 − 55 = 80.

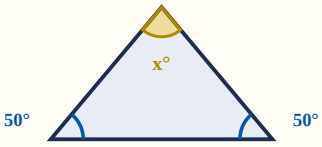
★ Try it **7) Exterior Angle Theorem, remote 46°, exterior = 118°. x = ?**



x = 72°

How to do it: Exterior angle = sum of the two far inside angles: x + 46 = 118, so x = 118 − 46 = 72.

★ Try it **8) Angle Sum, given 50°, 50°. x = ?**

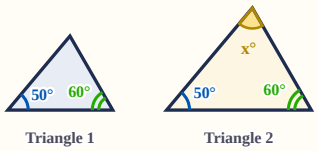


x = 80°

How to do it: The angles of a triangle add to 180°: x = 180 − 50 − 50 = 80.

D. Angle-Angle (AA) Similarity

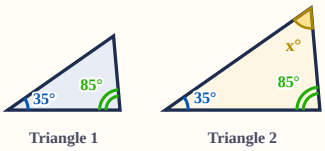
★★ On level **1) AA Similarity — two angles match: 50° and 60°. x = ?**



x = 70°

How to do it: Two angles match (50° and 60°), so the triangles are similar by AA. The third angle: x = 180 − 50 − 60 = 70.

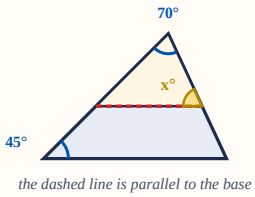
★★ On level **2) △ABC ~ △DEF, given 35° and 85°. x = ?**



x = 60°

How to do it: ∠C = 180 − 35 − 85 = 60°. ∠F corresponds to ∠C, so x = 60.

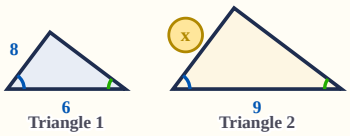
★★ On level **3) Parallel line inside a triangle, given 70° and 45°. x = ?**



x = 65°

How to do it: The dashed line is parallel to the base, so the small top triangle has the same angles as the big one (corresponding angles). Its three angles add to 180°: x = 180 − 70 − 45 = 65.

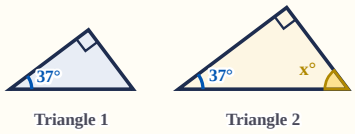
★★ On level **4) Similar triangles — sides 6 and 8 match 9 and x. x = ?**



x = 12°

How to do it: Scale factor = 9 ÷ 6 = 1.5. x = 8 × 1.5 = 12.

★★ On level **5) Similar right triangles, given 37°. x = ?**

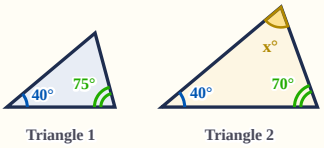


Triangle 1 Triangle 2

x = 53°

How to do it: Both triangles have a 90° angle and a 37° angle, so they are similar by AA. x = 180 − 90 − 37 = 53.

★★ On level **6) Triangle 1: 40°, 75°. Triangle 2: 40°, 70°. Find x, Triangle 2's third angle.**



Triangle 1 Triangle 2

x = 70°

How to do it: Triangle 2's angles add to 180°: x = 180 − 40 − 70 = 70. (Triangle 1's third angle is 180 − 40 − 75 = 65°, so the triangles are NOT similar.)